Individual Reflection Report

AI and Text Analytics Coursework – Spring 2026
University of Bristol
Team 16, Task 1: PubMedQA Biomedical Question Answering

Peidong Wang
hz25950@bristol.ac.uk

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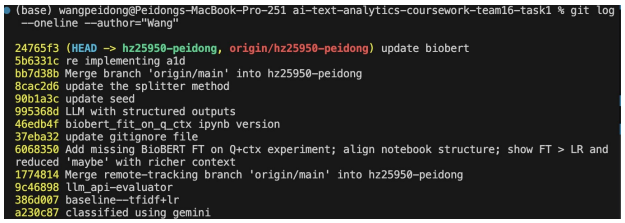
1 Contributions Summary

I led the **text-representation axis** (Axis 1), investigating how input encoding affects biomedical reasoning performance. My strategy was incremental: starting from sparse baselines, moving to frozen contextual embeddings, and concluding with full model fine-tuning. To ensure experimental rigour across the team, I also developed a custom stratified splitting script (`data_splitter.py`) for robust 10-fold cross-validation.

My concrete contributions were:

- **TF-IDF + LR Baselines (A1a/b/c):** Implemented three input variants (Q, Q+Ctx, Q+Ctx+Ans) to establish sparse performance ceilings (`baseline_lr_group.py`).
- **BioBERT Embeddings + LR (A1d):** Extracted frozen representations from `biobert-v1.2` to evaluate contextual embeddings for linear classification.
- **BioBERT Fine-tuning (A1e/f):** Adapted the full model with weighted cross-entropy for Q+Ctx and oracle Q+Ctx+Ans inputs (`modeling_biobert_ft_*.py`).



```
(base) wangpeidong@Peidongs-MacBook-Pro-251 ai-text-analytics-coursework-team16-task1 % git log
  --oneline --author="Wang"

24765f3 (HEAD -> hz25950-peidong, origin/hz25950-peidong) update biobert
5b6331c re implementing a1d
bb7d38b Merge branch 'origin/main' into hz25950-peidong
8cac2d6 update the splitter method
90b1a3c update seed
995368d LLM with structured outputs
46edb4f biobert_fit_on_q_ctx ipynb version
37eba32 update gitignore file
6068350 Add missing BioBERT FT on Q+ctx experiment; align notebook structure; show FT > LR and
reduced 'maybe' with richer context
1774814 Merge remote-tracking branch 'origin/main' into hz25950-peidong
9c46898 llm_api-evaluator
386d007 baseline--tfidf+lr
a230c87 classified using gemini
```

Figure 1: *

Figure 1: GitHub commit activity – commits attributed to Peidong Wang.

2 Technical Understanding

2a. Technical Challenge: Context Window Optimization (A1d)

A significant challenge was that frozen BioBERT embeddings (A1d) initially underperformed sparse baselines. I diagnosed this as a **context window limitation**: the default sentence-transformer configuration truncated inputs at 100 tokens, losing critical evidence. I re-implemented the extraction pipeline using `AutoModel` with a 384-token window and mean-pooling. This optimization raised test macro-F1 from 0.353 to 0.419 (+18.7%), demonstrating that representation quality was limited by encoding parameters rather than model expressiveness. Additionally, I implemented weighted cross-entropy to handle the extreme class imbalance (11% *maybe* class).

2b. Impact on Group Results

My Axis-1 experiments provided the team's central empirical finding: while sparse models plateau regardless of input richness, transformer models can leverage extended context (A1f), achieving 0.466 macro-F1. This established the "representation matters" narrative and motivated our final transition to PubMedBERT (A1g), which yielded our peak result of 0.604. My work ensured that the baseline and intermediate transformer experiments were statistically robust and properly weighted.

2c. Reflection and Future Work

Exploratory zero-shot testing with the Gemini API achieved 0.611 macro-F1, surpassing our fine-tuned models. This indicates that future work should explore generative reasoning (e.g., Chain-of-Thought) to improve evidence detection for the *maybe* class,

moving beyond the representation-focused scope of Axis 1.

3 Teamwork

The team coordinated via **GitHub**, **WhatsApp**, and **Jira**, with weekly **Microsoft Teams** meetings to resolve blockers. To ensure inclusion and consistency, all experimental results were cross-checked by teammates.

Role assignments were as follows:

- **Peidong Wang** – I led the **text-representation axis (Axis 1)**, developing the TF-IDF baselines and BioBERT fine-tuning pipelines. My focus was on optimizing input composition (e.g., adding long-answer channels), tuning the encoding context window, and implementing weighted loss functions to address the *maybe*-class imbalance.
- **Abdullah** – Data loading, preprocessing, and 10-fold CV infrastructure.
- **Zhengjie** – Axis-2 classical models and ensemble methods.
- **Abhigyan** – Error analysis and misclassification inspection.
- **Tanishk** – Advanced transformer models (PubMedBERT) and final integration.

A shared random seed (`SEED=42`) and a unified data-splitting protocol were adopted by all members to guarantee reproducibility.